



Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: II Semester B.E. ECE B

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member: P.Venkatesh

Innovative Practice Description

- **Unit / Topic:** Unit IV / TRANSIENTS AND RESONANCE IN RLC CIRCUITS
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO22
- **Activity Chosen:** Team Assisted Individualization (TAI)
- **Justification:**

Team Assisted Individualization (TAI) is used to improve the academic behaviour of the students, to increase their Problematic knowledge more than traditional methods. There are three major impedance matching techniques are available. Impedance matching networks can be designed by analytical method and graphical method (Smith Chart). In analytical method, conventional design equations are used to design matching networks. So, TAI method can be used to enrich the analytic knowledge of the students.

- **Time Allotted for the Activity:** One Period (50 Minutes)

- **Details of the Implementation:**

At first the students are divided into groups of four or 5 members (10 Groups). Discussed the Team Assisted Individualization (TAI) activity to the students. TAI activity objectives, outcomes, rubrics are explained and provided problems to be solved.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PSO3
CO3	3	2	2	2	2	2	3	3

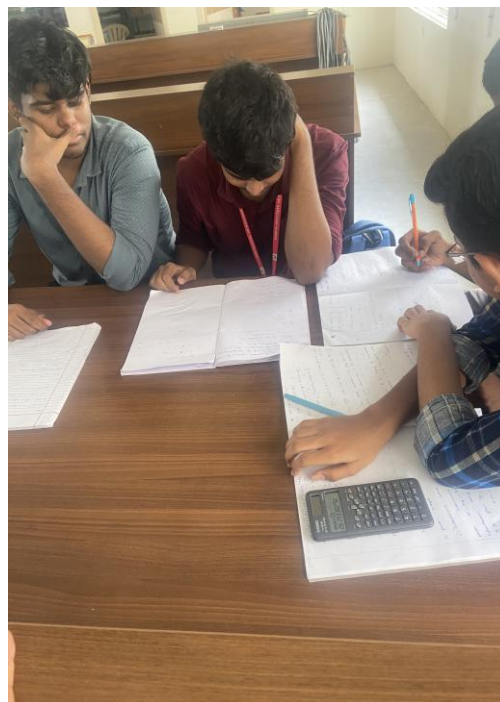
- **PO / PSO mapped:**

Innovative practice	PO4	PO8	PO9	PO10	PSO3
	2	2	2	3	3
Justification for correlation	The course outcome is	Identify tenets of the	A collaborative	The topics deals analysis	The concept of RLC networks

	moderately correlated since it involves the analysis of different lumped components	professional code of ethics while doing activity-based learning. So, it is moderately correlated	activity TAI is planned which involves Implement the norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work, to accomplish a goal. This course outcome is correlated moderately.	of RLC networks involves complement writing and presentations. This task is accomplished through assignment therefore course outcome is correlated highly.	are required for design, analyze communication systems. So, the course outcome is highly correlated.
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- Images / Screenshot of the practice:

Single stub matching using Smith chart - Team Assisted Individualization (TAI)



- **Reflective Critique:**
- ❖ ***Feedback of practice from students and other stakeholders:***
Students feel that it was interesting. They learn how to communicate with team members and work together.
- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
Before taking this activity, students felt difficult to understand the problems of impedance matching networks. After completing this, they have rectified their doubts within that group. They have solved the impedance matching problems effectively in a group.
- ❖ ***Challenges faced in implementation:***
Most of the student groups did the problems effectively. However, some of the student groups had different thoughts while solving the problems and also they have not communicated effectively with in that group. Due to that some minor errors are happened at the time of problem solving.

References:

- ❖ Slavin, R. E., Madden, N. A., & Leavey, M. (1984). Effects of team assisted individualization on the mathematics achievement of academically handicapped and nonhandicapped students. *Journal of Educational Psychology*, 76(5), 813–819.
<https://doi.org/10.1037/0022-0663.76.5.813>
- ❖ <https://eric.ed.gov/?id=ED232852>
- ❖ <https://psycnet.apa.org/record/1985-10827-001>

Signature of Faculty Member

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Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: II semester B.E. ECE B

Course Code & Title: EC3251 Circuit Analysis

Name of the Faculty member (s): P. Venkatesh

Innovative Practice Description

- **Unit / Topic:** Unit V / COUPLED CIRCUITS AND TOPOLOGY
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO28
- **Activity Chosen:** Flipped Classroom
- **Justification:**

It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning. Also the practice was chosen because the topic is simpler and interesting for the students to refer and learn by themselves

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Apart from this they can refer any other relevant source for preparation. Students are asked to prepare more in depth than before. Students are separated into groups where students are given a specific topic to prepare and present. Get the class back together to share the individual group's work with everyone.

- **CO – PO / PSO mapping:**

CO	PO1	PO3	PO8	PO9	PO10
CO5	3	2	1	1	1

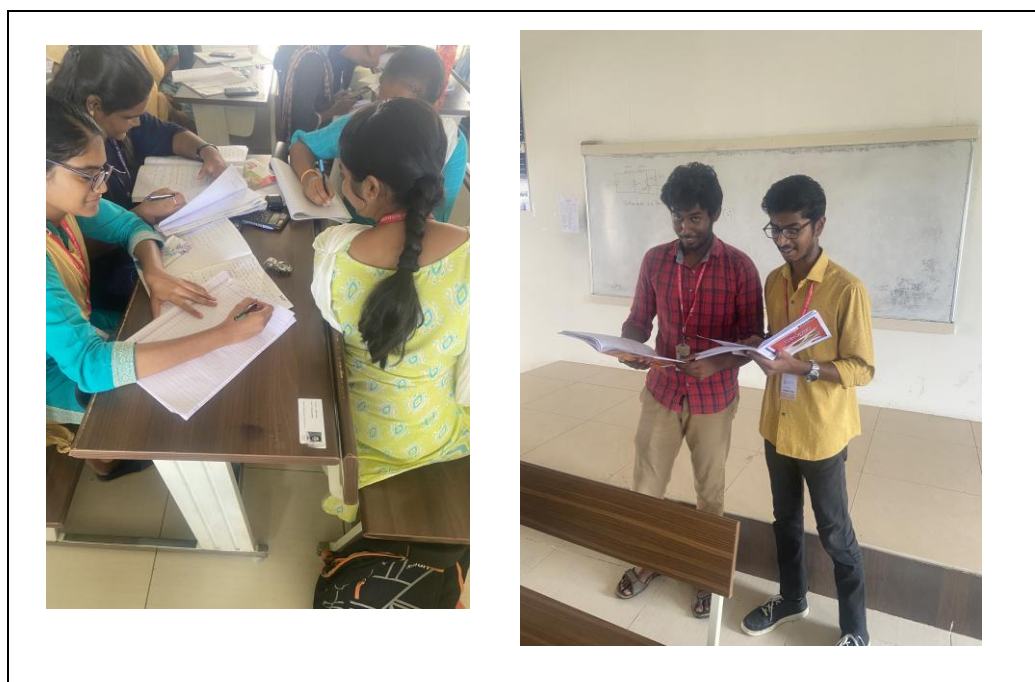
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO8	PO9	PO10
	3	2	1	1	1
Justification for	Solve problems in	PO3 is moderately mapped with	Identify tenets of the	It helps the students to	The students will present the

correlation	topics nodal analysis using cut set and hence PO1 has been mapped strongly to level 3	level because students will build new method for current analysis using cut set schedule	2 professional code of ethics while doing activity based learning. So, it is mapped at level 2	present the topic of discussion as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	results and conclusions of the activity carried out in front of the class which require good communication and presentation skills and so PO 10 is mapped
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• **Images / Screenshot of the practice:**



• **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***
 - Students feel that they have improved self-learning.
 - They learn how to communicate with team members and work together.
 - They were completely involved
- ***Benefit of the practice:***
 - Every student got equal opportunity to come forward to take part in this activity.
 - They felt easy in solving level 3 questions in Assignments and Internal Assessment test

- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct
- ***Challenges faced in implementation:***
 - Few students hesitated to come forward to present the results and conclusions of the discussion
 - The main challenge faced is that few students not exposed to flipped class room.

References:

- ❖ <https://www.teachthought.com/learning/a-flipped-classroom/>
- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.



Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: I semester B.E. ECE B

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): P.Venkatesh

Innovative Practice Description

- **Unit / Topic:** V / Hyper parameter Tuning
- **Course Outcome:** CO5
- **Topic Learning Outcome:** 5d
- **Activity Chosen:** Jigsaw
- **Justification:**

The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal each person depends on all the others.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10
CO1	2	3	2	3	3

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO8	PO9	PO10
	2	3	2	3	3
Justification for correlation	Hyper parameter Tuning requires the knowledge of engineering	The course outcome is highly correlated, since identification of	Identify tenets of the professional code of ethics while doing activity-	A Collaborative activity – Jigsaw helps the students to Present results as a	The topics deals with presentations of fixed- and floating-point representation. This task is

	fundamentals such as representation of numbers and the order. This course outcome is moderately correlated.	mathematical and engineering knowledge to solve problems related to hyper parameter tuning	based learning. So, it is moderately correlated.	team, with smooth integration of contributions from all individual efforts. The course outcome is highly correlated.	accomplished through Jigsaw activity therefore course outcome is correlated highly.
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- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students feel that it was interesting. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ https://www.educationworld.com/a_curr/strategy/strategy036.shtml
<https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>